A Descriptive Analysis of AI Adoption, Work Models, and Compensation in the Software Industry

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I. Introduction

A. Overview

The software engineering landscape is undergoing rapid transformation driven by continuous technological advancements. As millions of lines of code are deployed daily, understanding the developers behind this production becomes increasingly important. The 2024 Stack Overflow Developer Survey offers a comprehensive snapshot of this global workforce, with responses from over 65,000 participants across 185 countries.

Rather than focusing solely on technical outputs, this study aims to explore the human and structural dimensions of the software industry. In particular, it attempts to capture how developers' backgrounds, tools, and working environments interact within a data-driven framework.

B. Motivation

The traditional computer science degree is no longer the only pathway into the software industry, as self-taught developers now represent a significant portion of the workforce. At the same time, AI-assisted development tools have started to reshape how coding tasks are approached on a daily basis.

This raises several interesting questions. For instance, do more experienced developers actually adopt these tools more frequently, or are they primarily embraced by newer entrants? Similarly, it is not immediately clear whether the use of AI tools translates into measurable financial benefits.

Motivated by these uncertainties, this study investigates how educational background, professional experience, and work models relate to both AI adoption and salary outcomes.

C. Research Questions

To guide this descriptive analysis, the following research questions (RQs) were established:

- **RQ1:** Does formal university education create a significant difference in a developer's long-term industry survival compared to alternative educational paths?
- **RQ2:** How does a developer's level of professional experience relate to their likelihood of adopting AI-based coding tools?
- **RQ3:** Is there a noticeable difference in annual compensation between developers who actively integrate AI into their workflows and those who do not?
- **RQ4:** How do different work environments (Remote, Hybrid, In-person) influence the distribution of developer salaries?
- **RQ5:** Which popular programming language communities exhibit the highest rates of AI tool adoption?

II. Method

A. Dataset

1) Overview: The dataset used in this study is derived from the May 2024 Stack Overflow Developer Survey. To enable quantitative analysis, categorical and textual responses were encoded into numerical representations (binary encoding and label encoding where appropriate).

During preprocessing, records with missing values in key variables were removed. Additionally, extreme salary outliers were filtered using a range-based approach, restricting annual compensation values to between $10,000 and $500,000. While

somewhat arbitrary, this range was chosen to exclude unrealistic or highly noisy entries without overly distorting the distribution.

After cleaning, a total of 27,954 valid observations remained for analysis. But one limitation of this analysis is that the dataset relies on self-reported survey responses, which may introduce bias.

2) Attributes: The specific variables selected to address the research questions are defined as follows:

- **EdLevel_Numeric:** A binary indicator of formal education (1: University degree, 0: Self-taught or alternative).
- **YearsCodePro:** A continuous numerical variable representing total years of professional coding experience.
- **AISelect_Numeric:** A binary flag indicating active usage of AI development tools (1: Uses AI, 0: Does not use AI).
- **ConvertedCompYearly:** A continuous variable tracking annual financial compensation in USD.
- **DevType:** A categorical attribute specifying the respondent's primary professional role.
- **LanguageHaveWorkedWith:** A textual variable listing the programming languages used by the developer.
- **RemoteWork:** A categorical variable denoting the current work environment (Remote, Hybrid, or In-person).

III. Results

A. Descriptive Analysis

A set of descriptive statistical techniques was applied to explore the underlying structure of the dataset. Measures such as mean, median, variance, and standard deviation were computed, followed by visual analysis.

The main observations can be summarized as follows:

- **Boxplots** comparing professional experience across educational backgrounds suggest that formal university education does not create a substantial difference in long-term industry presence. Interestingly, the spread of experience appears relatively similar across both groups, which may indicate that alternative learning paths can be equally sustainable.
- A **Histogram** combined with Kernel Density Estimation (KDE) shows that annual salaries follow a strongly right-skewed distribution. This is consistent with typical income distributions, where a large portion of individuals cluster around moderate salaries, while a smaller group earns significantly higher amounts.
- **Bar chart** comparisons indicate that developers who do not use AI tools tend to have slightly higher average salaries. However, this difference should be interpreted cautiously; it is likely influenced by confounding factors such as seniority, where highly experienced developers (who generally command higher salaries) may adopt new AI tools at a slower rate.
- The distribution of developer roles via a **Pie Chart** highlights the dominance of a few key positions within the dataset, suggesting that the results are shaped primarily by these major groups.
- A **Horizontal Bar Chart** of AI adoption across programming languages reveals noticeable variation between communities. A possible explanation is that certain ecosystems may integrate AI tools more naturally into their workflows.
- Finally, salary distributions across work models via a second **Boxplot** show some variation between remote, hybrid, and in-person roles. While remote positions demonstrate noticeably higher median salaries and upper quartiles, the overall distributions still overlap considerably.

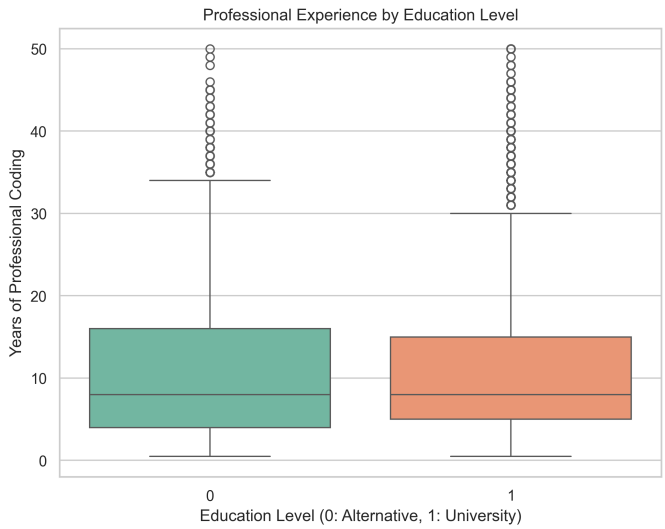

Fig. 1. Distribution of professional experience years according to educational backgrounds.

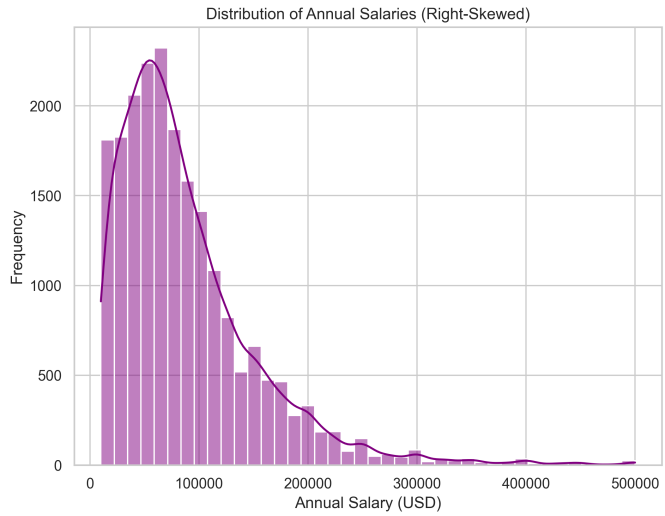

Fig. 2. Right-skewed distribution of annual salaries (USD).

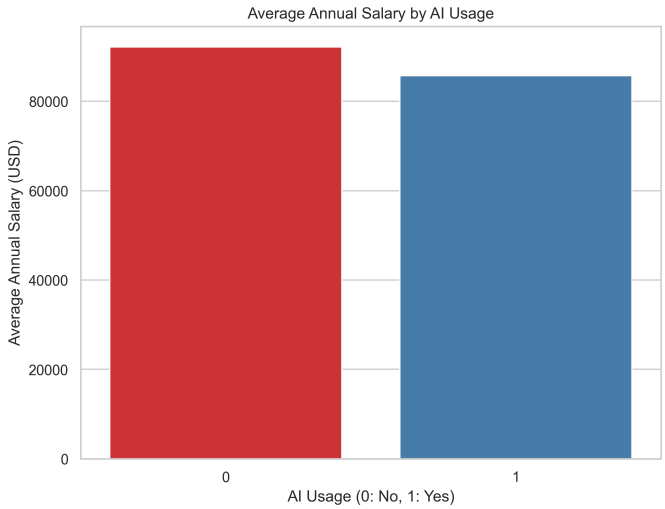

Fig. 3. Average annual salary comparison based on AI usage.

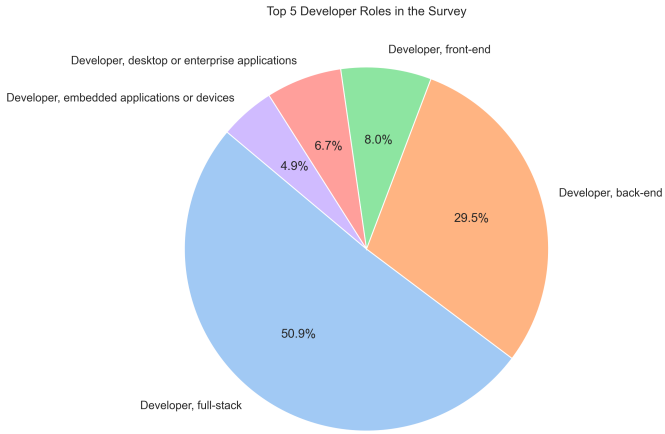

Fig. 4. Distribution of the top 5 developer roles.

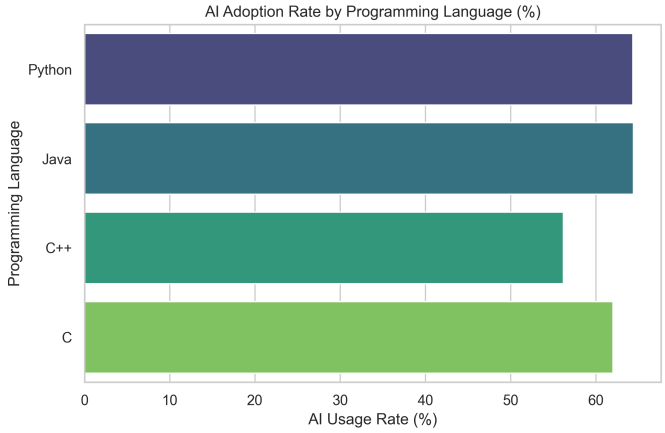

Fig. 5. AI adoption rates among popular programming languages.

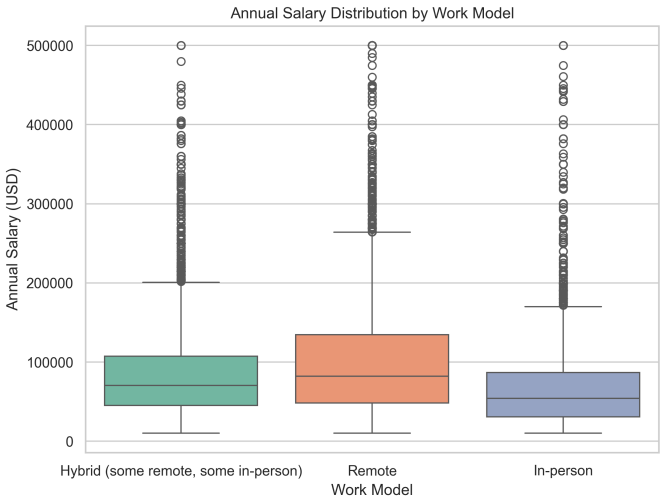

Fig. 6. Annual salary distribution compared by work models.

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